04 — Multimodal Text & Image Fusion

Goal: Combine state-of-the-art text (TF-IDF + SVM, CamemBERT) and image models (ConvNeXt-Tiny) to maximise classification performance.

- **Image Model**: `ConvNeXt-Tiny` (Full Fine-Tuning + Smart Cropping from `notebooks/03_image_fineTuning/03_7_image_convnext_optimized.ipynb`), reaching **0.6998 Weighted F1** alone.
- **Text Models**: Calibrated `LinearSVC` (**0.8430 Weighted F1**) and fine-tuned `CamemBERT` (**0.8765 Weighted F1**).
- **Fusion Strategies**:
  1. **Soft Voting**: Probability average of the models. Equal weights on the three models, and weighted soft voting where the weights are chosen by cross-validation (SVM + image, CamemBERT + image, CamemBERT + SVM + image).
  2. **Stacking Classifier (Random Forest)**: Meta-classifier trained via 5-fold cross-validation on the combined 81-dimensional probability representations (27 classes × 3 models).

1. Text probabilities

We use the optimised text model from notebook 03_2 (all features, binary TF-IDF, LinearSVC with C = 1). A LinearSVC gives scores, not probabilities. Calibration learns on the training set (3-fold CV) how to turn these scores into probabilities.

```
Text probabilities saved: (16984, 28)
Each row sums to 1      : True
```

2. Match text and image on the same products

```
Products scored by both: 16984
```

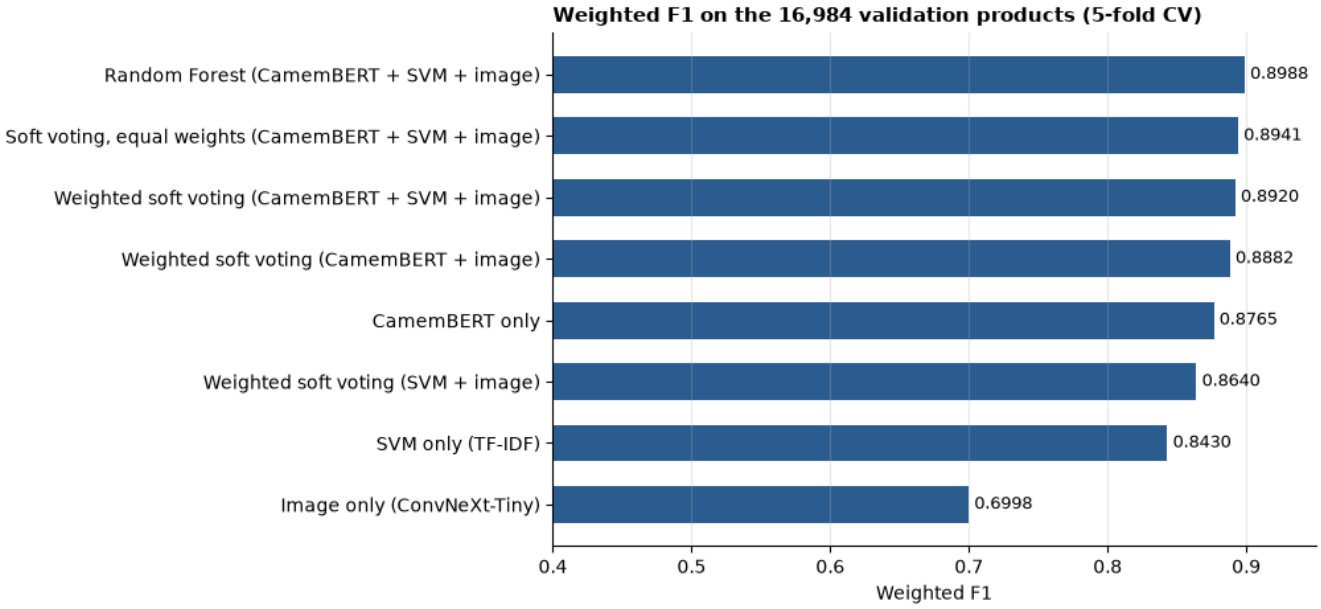
3. Combine the models

Text and image probabilities exist for the 16984 validation products, which none of the models were trained on. We score every method with 5-fold cross-validation on these products: the combiner learns on 4/5 of them and predicts the remaining 1/5. Every product is predicted by a combiner that never saw it.

Two ways to combine:

- Soft voting: average the probabilities of the models, then take the class with the highest average.
  - Equal weights: each of the three models counts 1/3. Nothing is chosen.
  - Weighted soft voting: each model gets its own weight (steps of 0.1, summing to 1). The weights are chosen on the 4/5 part only.
- Random Forest: a model that learns from the 81 probabilities (27 classes × 3 models) how to combine them.

```
Weighted soft voting (SVM + image) — weights chosen in each fold: [(0.6, 0.4), (0.6, 0.4), (0.6, 0.4), (0.6, 0.4), (0.6, 0.4)]
Weighted soft voting (CamemBERT + image) — weights chosen in each fold: [(0.5, 0.5), (0.5, 0.5), (0.5, 0.5), (0.5, 0.5), (0.5, 0.5)]
Weighted soft voting (CamemBERT + SVM + image) — weights chosen in each fold: [(0.4, 0.3, 0.3), (0.4, 0.3, 0.3), (0.3, 0.4, 0.3), (0.4, 0.3, 0.3), (0.4, 0.3, 0.3)]
```



Model	Weighted F1	Macro F1
Random Forest (CamemBERT + SVM + image)	0.8988	0.8856
Soft voting, equal weights (CamemBERT + SVM + image)	0.8941	0.8826
Weighted soft voting (CamemBERT + SVM + image)	0.8920	0.8797
Weighted soft voting (CamemBERT + image)	0.8882	0.8730
CamemBERT only	0.8765	0.8594
Weighted soft voting (SVM + image)	0.8640	0.8548
SVM only (TF-IDF)	0.8430	0.8326
Image only (ConvNeXt-Tiny)	0.6998	0.6752

Analysis

- The best model is the Random Forest on CamemBERT, SVM and image: 0.8988 weighted F1 and 0.8856 macro F1.
- That is +2.2 points over CamemBERT alone (0.8765) and +5.6 points over the SVM alone (0.8430).
- Soft voting with equal weights on the three models reaches 0.8941. It already gives most of the gain.
- Weighted soft voting on the same three models reaches 0.8920 (weights 0.4 CamemBERT, 0.3 SVM, 0.3 image in 4 of 5 folds). Choosing the weights does not help: equal weights are 0.2 points better.
  - The weights that score best on the 4/5 part do not transfer exactly to the remaining 1/5.
  - Equal weights (1/3) are not in the 0.1 grid. The closest options are 0.4/0.3/0.3 and 0.3/0.4/0.3.
- Weighted soft voting with CamemBERT and image (0.5 each) reaches 0.8882. With SVM and image (0.6 SVM, 0.4 image) it reaches 0.8640.
- The weights are almost the same in every fold, so the choice is stable.
- We also tried Gradient Boosting instead of Random Forest (appendix notebook). It scored slightly lower and was much slower, so we kept the Random Forest.

4. Per category: best fusion vs CamemBERT

Code	Category	CamemBERT F1	Best fusion F1	Gain
2705	New Books	0.818	0.926	0.108
10	Used Books	0.689	0.788	0.099
1180	Wargames	0.614	0.706	0.093
40	Games	0.795	0.855	0.059
1302	Outdoor Toys	0.810	0.853	0.043
1140	Figurines	0.834	0.862	0.028
60	Consoles	0.876	0.903	0.027
1940	Food	0.946	0.969	0.023
1160	Cards	0.955	0.978	0.023
1320	Baby	0.856	0.877	0.021
2585	Garden Tools	0.823	0.843	0.019
2403	Book Lots	0.868	0.886	0.018
2462	Used Games	0.821	0.838	0.017
2220	Pets	0.890	0.906	0.017
2582	Garden Furniture	0.794	0.810	0.016
50	Game Accessories	0.880	0.896	0.016
1281	Board Games	0.684	0.700	0.015
2522	Stationery	0.945	0.958	0.014
2583	Pool	0.970	0.983	0.013
1920	Textiles	0.925	0.938	0.013
1280	Toys	0.791	0.803	0.012
1560	Furniture	0.882	0.890	0.008
2060	Decor	0.861	0.867	0.006
2280	Magazines	0.919	0.922	0.003
1300	RC Models	0.987	0.987	0.000
2905	PC Games	0.991	0.991	0.000
1301	Baby Socks	0.978	0.975	-0.003

Analysis

- The best fusion is better than CamemBERT on 26 of the 27 categories. Only Baby Socks drops slightly (-0.3 points).
- The largest gains are on New Books (+10.8 points), Used Books (+9.9) and Wargames (+9.3).
- Books gain the most from the image: the cover shows if a book is new, used, or a lot of several books.
- Wargames was CamemBERT's weakest category. The SVM catches its rare exact words (*warhammer*, *heroclix*), so combining both fixes it.

5. Best fusion vs weighted soft voting

The Random Forest against the best weighted soft voting (CamemBERT + SVM + image), per category, on the same 5-fold predictions.

```
Errors: weighted soft voting 1813, Random Forest 1715 (98 fewer)
Categories where the Random Forest is better: 18 of 27
```

Code	Category	Weighted soft voting	Random Forest	Gain
2705	New Books	0.884	0.926	0.042
10	Used Books	0.757	0.788	0.031
60	Consoles	0.885	0.903	0.019
1281	Board Games	0.684	0.700	0.016
1300	RC Models	0.973	0.987	0.014
1280	Toys	0.789	0.803	0.014
40	Games	0.841	0.855	0.014
1920	Textiles	0.925	0.938	0.013
1940	Food	0.957	0.969	0.012
50	Game Accessories	0.886	0.896	0.010
1302	Outdoor Toys	0.843	0.853	0.010
2583	Pool	0.975	0.983	0.009
2060	Decor	0.862	0.867	0.005
1160	Cards	0.973	0.978	0.005
1301	Baby Socks	0.972	0.975	0.003
2280	Magazines	0.922	0.922	0.001
2522	Stationery	0.958	0.958	0.001
2403	Book Lots	0.886	0.886	0.000
1560	Furniture	0.891	0.890	-0.001
2905	PC Games	0.994	0.991	-0.003
2585	Garden Tools	0.847	0.843	-0.004
2220	Pets	0.911	0.906	-0.005
1140	Figurines	0.868	0.862	-0.006
1320	Baby	0.885	0.877	-0.008
1180	Wargames	0.715	0.706	-0.009
2462	Used Games	0.848	0.838	-0.010
2582	Garden Furniture	0.822	0.810	-0.012

Analysis

- The Random Forest makes 1715 errors against 1813 for weighted soft voting (98 fewer). Weighted F1: 0.8988 against 0.8920.
- It is better on 18 of the 27 categories. The largest gains are on New Books (+4.2 points), Used Books (+3.1) and Consoles (+1.9).
- These are the categories where one model should be trusted more than the others. For books, the image tells new from used. Soft voting uses the same weights for every category, so it cannot do this.
- On 9 categories, weighted soft voting is slightly better, by at most 1.2 points (Garden Furniture). Wargames, Used Games and Baby lose about 1 point.
- So the Random Forest gains most of its lead on a few categories, mainly books.

Conclusion

- Combining models helps. Soft voting on the three models already reaches 0.8941 with equal weights. Choosing the weights does not improve it.
- A learned combination (Random Forest) adds 0.5 points more, mainly on the book categories.
- Best model: Random Forest on the CamemBERT, SVM and ConvNeXt probabilities, with 0.8988 weighted F1 on the validation products (5-fold CV).
- Each part adds something: CamemBERT reads context, the SVM catches rare specific words, and the image helps on books.